

Geometry

Unit 11

Lesson 1 Homework

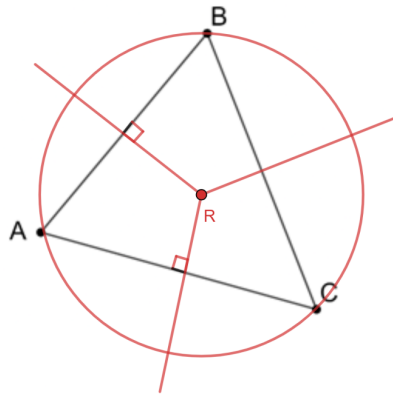
Name _____

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Use $\triangle ABC$ to answer questions 1-7.

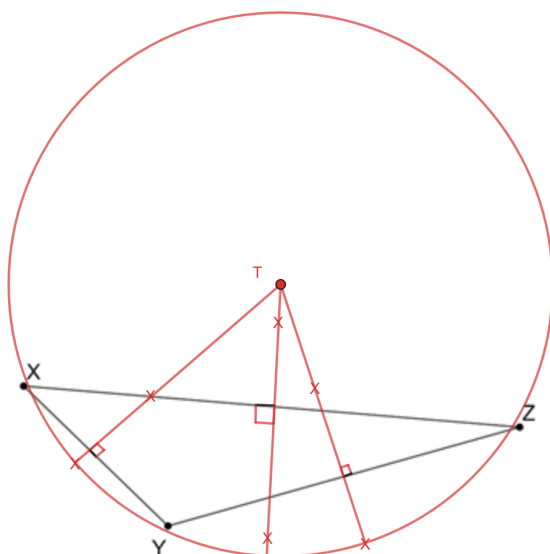
1. Construct the three perpendicular bisectors for $\triangle ABC$.



2. Label the circumcenter of the triangle as Point R .
3. Use a ruler to measure the length of segment AR .
 2.8 cm
4. Use a ruler to measure the length of segment CR .
 2.8 cm
5. Use a ruler to measure the length of segment BR .
 2.8 cm
6. Use a compass to draw the circumscribed circle around the triangle.

Use $\triangle XYZ$ to answer questions 7-10.

Commented [JP1]: <https://www.geogebra.org/calculator/>

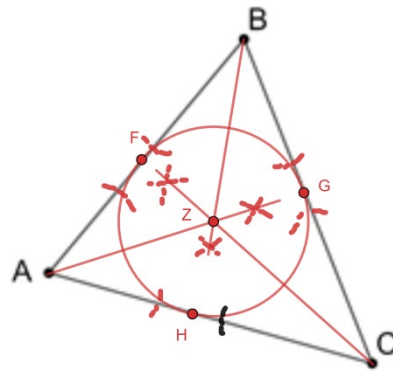


7. Construct the three perpendicular bisectors for $\triangle XYZ$.
8. Label the circumcenter of the triangle as Point T .
9. Use a compass to draw the circumscribed circle.
10. Given that the measure of arc $XYZ = 140^\circ$, what is the measure of $\angle Y$?

$$m\angle = 110^\circ$$

Use $\triangle ABC$ to answer questions 1-5.

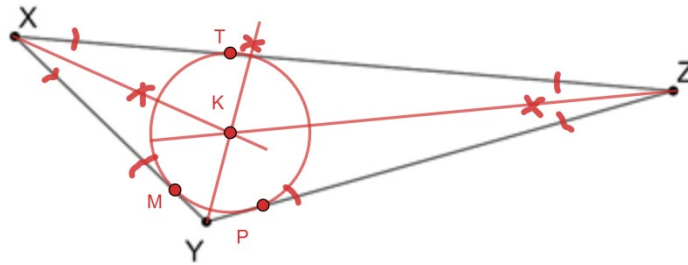
- Construct the three angle bisectors for $\triangle ABC$.



- Label the incenter of the triangle as Point Z .
- Use a compass to draw the inscribed circle inside the triangle, with point Z as its center.
- Label the three points where the inscribed circle intersects the triangle as F , G , and H .
- Measure the lengths of segments $\overline{FZ}$, $\overline{GZ}$, and $\overline{HZ}$.
Answers may vary based on unit of measure, where $FZ = GZ = HZ$.

Use $\triangle XYZ$ to answer questions 6-10.

6. Construct the three angle bisectors for $\triangle XYZ$.

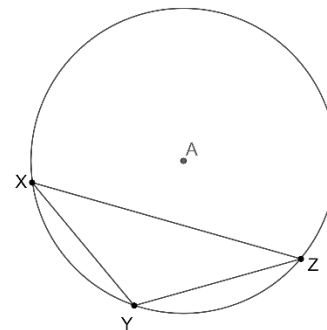


7. Label the incenter of the triangle as Point K .
8. Use a compass to draw the inscribed circle inside the triangle, with point K as its center.
9. Label the three points where the inscribed circle intersects the triangle as M , P , and T .
10. Measure the lengths of segments $\overline{MK}$, $\overline{PK}$, and $\overline{TK}$.
Answers may vary based on unit of measure, where $MK = PK = TK$.

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Lesson 3 Homework

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Circle A is shown, where points X, Y , and Z are on circle A and create $\triangle XYZ$. $m\widehat{XZ} = 240^\circ$, $m\angle ZXY = 20^\circ$, and $m\angle XYZ = (4x - 28)^\circ$.



1. Find the value of x .

$$\begin{aligned} 2(4x - 28) &= 240 \\ 4x - 28 &= 120 \\ +28 &= +28 \\ 4x &= 148 \\ \frac{4x}{4} &= \frac{148}{4} \\ x &= 37 \end{aligned}$$

2. Find the $m\angle XYZ$.

$$\frac{240}{2} = 120^\circ$$

3. Find the $m\widehat{YZ}$.

$$2(20) = 40^\circ$$

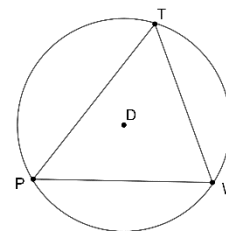
4. Find $m\angle YZX$.

$$20 + 120 + m\angle YZX = 180$$

$$140 + m\angle YZX = 180$$

$$m\angle YZX = 40^\circ$$

Circle D is shown, where points P, T , and W are on circle D and create $\triangle PTW$. $m\angle PTW = 54^\circ$ and $m\angle TWP = 60^\circ$.



5. Find $m\widehat{PT}$.

$$2(60) = 120^\circ$$

6. Find $m\widehat{TW}$.

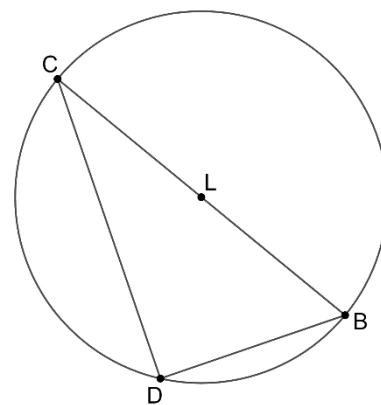
$$54 + 60 + m\angle P = 180$$

$$114 + m\angle P = 180$$

$$m\angle P = 66^\circ$$

$$2(66) = 132^\circ$$

Circle L has points B, C , and D on circle L and diameter $\overline{BC}$. $\triangle BCD$ is inscribed in the circle. $m\widehat{CD} = 100^\circ$, $m\angle DCB = 40^\circ$, and $m\angle DBC = (2x + 12)^\circ$.



7. What is the $m\angle CDB$?

90°

8. What is the value of x ?

$$2x + 12 = 50$$

$$-12 = -12$$

$$\frac{2x}{2} = \frac{38}{2}$$

$$x = 19$$

9. Determine $m\angle DBC$.

$$\frac{100}{2} = 50^\circ$$

10. Determine $m\widehat{DBC}$?

$$180 + 80 = 260^\circ$$

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Lesson 4 Homework

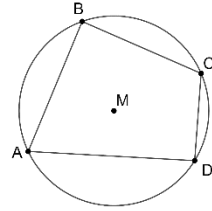
Name **Answer Key**

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Period _____

Circle M is shown, with $ABCD$ inscribed in the circle.

The $m\widehat{ADC} = 184^\circ$, and $m\angle DAB = 44^\circ$.



1. Find the $m\angle ABC$.

$$\frac{184}{2} = 92^\circ; m\angle ABC = 92^\circ$$

2. Find the $m\angle BCD$.

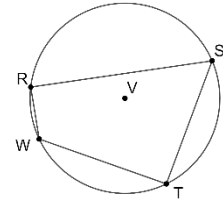
$$m\angle BCD = 180 - 44 = 136^\circ$$

3. Find the $m\angle CDA$.

$$m\angle CDA = 180 - 92 = 88^\circ$$

Circle V is shown, with $RSTW$ inscribed in the circle.

The $m\widehat{WTS} = 190^\circ$, and $m\angle WTS = (5x + 10)^\circ$.



4. Find the value of x .

$$\frac{190}{2} = 95^\circ; 180 - 95 = 85 = m\angle WTS$$

$$5x + 10 = 85$$

$$5x = 75$$

$$x = 15$$

5. Find the $m\angle WTS$.

$$m\angle WTS = 85^\circ$$

Circle B is shown, with $GHJK$ inscribed in the circle. The $m\angle HGK = (5x + 25)^\circ$, and $m\angle GKJ = (3x)^\circ$ and $m\angle KJH = (4x + 20)^\circ$.

6. Find the value of x .

$$5x + 25 + 4x + 20 = 180$$

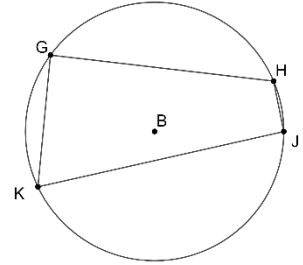
$$9x + 45 = 180$$

$$9x = 135$$

$$x = 15$$

7. Find $m\angle HGK$.

$$m\angle HGK = 5(15) + 25 = 100^\circ$$



8. Find $m\angle GKJ$.

$$m\angle GKJ = 3(15) = 45^\circ$$

9. Find $m\angle KJH$.

$$m\angle KJH = 4(15) + 20 = 80^\circ$$

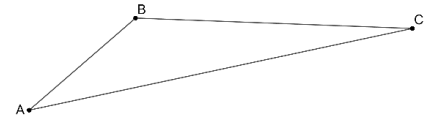
10. Find $m\angle JHG$.

$$m\angle JHG = 360 - (100 + 45 + 80) = 135^\circ$$

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Lesson 5 Homework

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For questions 1 through 4, $\triangle ABC$ is shown, where $m\angle A = (3x)^\circ$, $m\angle B = (7x + 10)^\circ$, and $m\angle C = (2x + 10)^\circ$.



1. Find the value of x .

$$\begin{aligned} 3x + (7x + 10) + (2x + 10) &= 180 \\ 12x + 20 &= 180 \\ 12x &= 160 \\ x &= 13\frac{1}{3} \end{aligned}$$

2. What is the $m\angle A$?

$$m\angle A = 3\left(13\frac{1}{3}\right) = 40^\circ$$

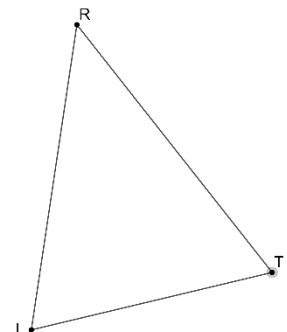
3. What is the $m\angle B$?

$$m\angle B = 7\left(13\frac{1}{3}\right) + 10 = 103\frac{1}{3}^\circ$$

4. What is the $m\angle C$?

$$m\angle C = 2\left(13\frac{1}{3}\right) + 10 = 36\frac{2}{3}^\circ$$

For questions 5 and 6, $\triangle LRT$ is shown, where $m\angle L = m\angle T = (5x)^\circ$ and $m\angle R = (2x)^\circ$.



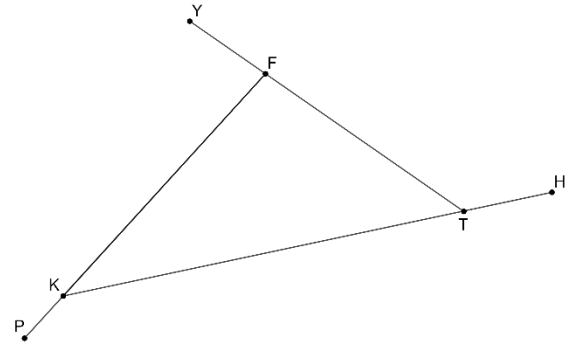
5. Find the value of x .

$$\begin{aligned} 5x + 5x + 2x &= 180 \\ 12x &= 180 \\ x &= 15 \end{aligned}$$

6. What is the $m\angle L$?

$$m\angle L = 5(15) = 75^\circ$$

For questions 7 through 10, $\triangle FKT$ is shown, where $\angle KFY$, $\angle PKT$, and $\angle FTH$ are exterior angles.
 $m\angle FTH = (8x + 10)^\circ$, $m\angle TFK = 88^\circ$, and
 $m\angle PKT = (16x - 30)^\circ$.



7. Find the value of x .

$$\begin{aligned} m\angle KFY &= 180 - 88 = 92^\circ \\ 92 + (16x - 30) + (8x + 10) &= 360 \\ 24x + 72 &= 360 \\ 24x &= 288 \\ x &= 12 \end{aligned}$$

8. Find $m\angle KTF$.

$$m\angle KTF = 180 - [8(12) + 10] = 74^\circ$$

9. Find $m\angle PKT$.

$$m\angle PKT = 16(12) - 30 = 162^\circ$$

10. Find $m\angle YFK$.

$$m\angle YFK = 180 - 88 = 92^\circ$$

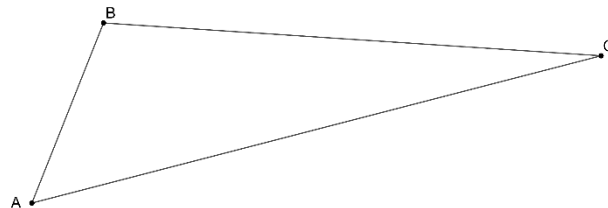
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Lesson 6 Homework

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1. $\triangle ABC$ is shown, where $AB = 4$, $BC = 8$, and $AC = 10$.

List the angles in order from least to greatest.

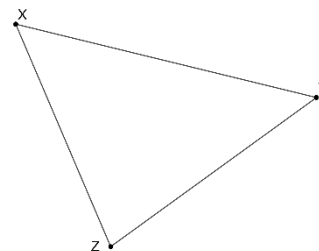
$\angle C, \angle A, \angle B$



$\triangle XYZ$ is shown. Use the triangle to complete questions 2 through 4.

2. Use a protractor to measure each of the angles.

$m\angle Z = 76^\circ$ $m\angle X = 53^\circ$ $m\angle Y = 51^\circ$



3. List the angles in order from smallest to largest.

$\angle Y, \angle X, \angle Z$

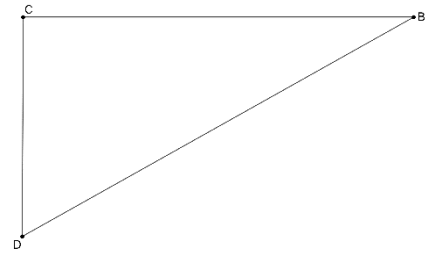
4. List the side lengths in order from smallest to largest.

$\overline{XZ}, \overline{YZ}, \overline{XY}$

5. $\triangle BCD$ is shown, where $m\angle B = 30^\circ$, $m\angle D = 60^\circ$, and $m\angle C = 90^\circ$.

List the side lengths in order from largest to smallest.

$\overline{BD}, \overline{BC}, \overline{CD}$



For questions 6 through 8, determine whether the given side lengths can form a triangle.

6. 12 cm, 14 cm, and 20 cm

$12 + 14 = 26; 26 > 20$ Yes it is a triangle.

7. 2 in, 3 in, and 5 in

$2 + 3 = 5; 5 = 5$ No, cannot be a triangle.

8. 3 ft, 8 ft, and 12 ft

$3 + 8 = 11; 11 < 12$ No, cannot be a triangle.

For questions 9 and 10, two sides of a triangle are given. Determine a possible side length for the third side.

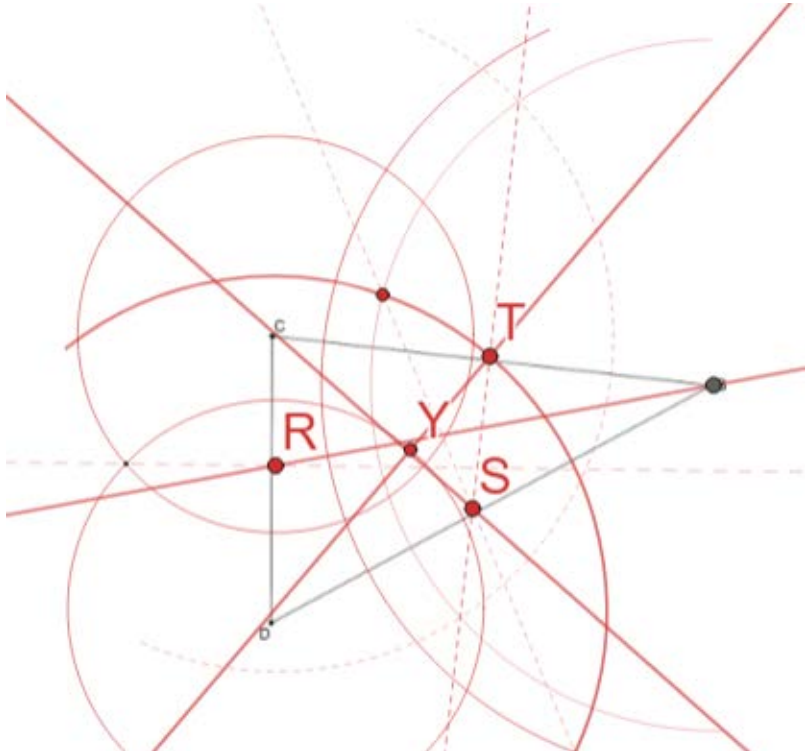
9. 7 in and 5 in

$7 + 5 = 12, 7 - 5 = 2$ Any value greater than 2 and less than 12.

10. 10 m and 2 m

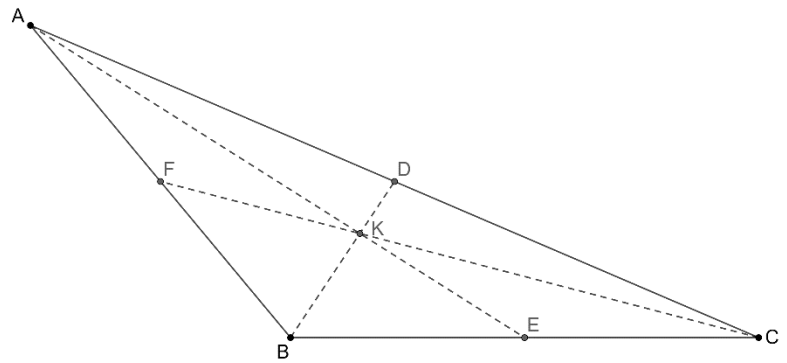
$10 + 2 = 12, 10 - 2 = 8$ Any value greater than 8 and less than 12.

$\triangle BCD$ is shown. Complete the steps for finding the centroid of the circle. Then, label the centroid Point Y .

Steps	Construction
Construct the perpendicular bisector of BC to find its midpoint, T .	
1. Construct the perpendicular bisector of $\overline{BD}$, label midpoint S .	
2. Construct the perpendicular bisector of $\overline{CD}$, label midpoint R .	
Draw the median, R , to the opposite vertex, $\angle DBC$.	
3. Draw the median, S , to the opposite vertex, $\angle DCB$.	
4. Draw the median, T , to the opposite vertex, $\angle BDC$.	
Draw Point Y , the point of concurrency, where the three medians intersect.	

Use this information to answer questions 6 through 8.

Triangle ABC is shown, where point K is the centroid of the triangle. $FC = 6$, $BK = 2.4$, and $KE = 3.7$.



6. Find the length of $\overline{DK}$.

$$2.4 \div 2 = 1.2$$

7. Find the length of $\overline{AE}$.

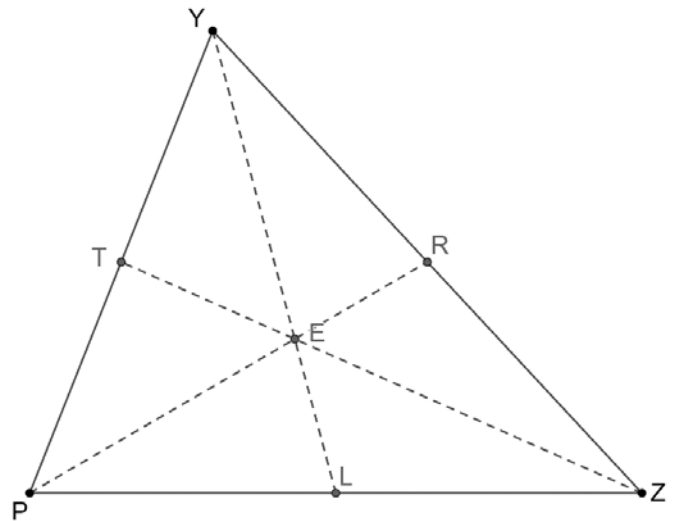
$$3.7 \cdot 3 = 11.1$$

8. Find the length of $\overline{KC}$.

$$\frac{2}{3} \cdot 6 = 4$$

Use this information to answer questions 9 and 10.

Triangle YPZ is shown, where point E is the centroid of the triangle. $TZ = 6x + 9$, and $TE = 4$.



9. Find the value of x .

$$6x + 9 = 12$$

$$6x = 3$$

$$x = \frac{1}{2}$$

10. Find the length of $\overline{EZ}$.

$$TZ = 6 \cdot \frac{1}{2} + 9 = 12$$

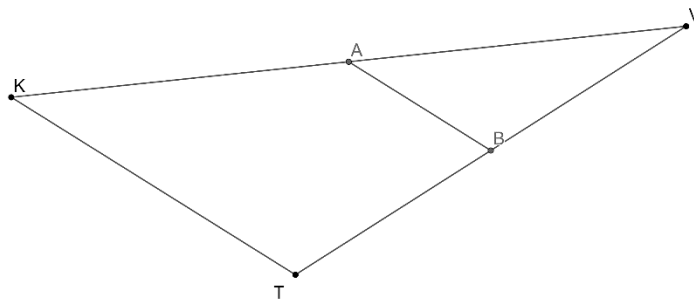
$$EZ = \frac{2}{3}(12) = 8$$

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Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{2}x + 3$, $AV = \frac{5}{2}y - 6.5$, $KA = \frac{1}{2}y + 3.5$, $VT = 3z - 7$, and $TB = 2x + 2.1$.



1. Find the value of x .

$$\begin{aligned} 2\left(\frac{1}{2}x + 3\right) &= 2x + 4 \\ x + 6 &= 2x + 4 \\ x &= 2 \end{aligned}$$

2. Find the value of y .

$$\begin{aligned} \frac{1}{2}y + 3.5 &= \frac{5}{2}y - 6.5 \\ 3.5 &= 2y - 6.5 \\ 2y &= 10 \\ y &= 5 \end{aligned}$$

3. Find the value of z .

$$\begin{aligned} 2(2x + 2.1) &= 3z - 7 \\ 2(2(2) + 2.1) &= 3z - 7 \\ 2(6.1) &= 3z - 7 \\ 12.2 &= 3z - 7 \\ 3z &= 19.2 \\ z &= 6.4 \end{aligned}$$

4. Find the length of the midsegment, $\overline{AB}$.

$$AB = \frac{1}{2}(2) + 3 = 4$$

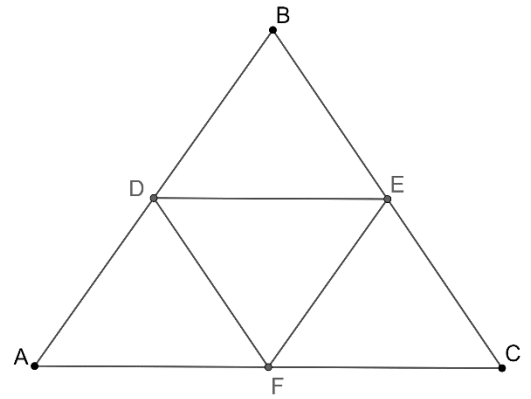
5. Find the length of $\overline{KV}$.

$$\begin{aligned} KV &= KA + AV \\ KV &= \frac{1}{2}(5) + 3.5 + \frac{5}{2}(5) - 6.5 = 12 \end{aligned}$$

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where D, E , and F are midpoints.

$DE = 2x - 4$, $AC = 4x - 8$, $DF = x + 3$, $BC = 3x + 1$,
 $FE = 3x - 6$, and $AB = 5x - 7$.



6. Find the value of x .

$$\begin{array}{ll} 2(x + 3) = 3x + 1 & 2(3x - 6) = 5x - 7 \\ 2x + 6 = 3x + 1 & 6x - 12 = 5x - 7 \\ x = 5 & x = 5 \end{array}$$

7. Find the length of $\overline{DF}$.

$$DF = 5 + 3 = 8$$

8. Find the length of $\overline{EF}$.

$$EF = 3(5) - 6 = 9$$

9. Find the length of $\overline{DE}$.

$$DE = 2(5) - 4 = 6$$

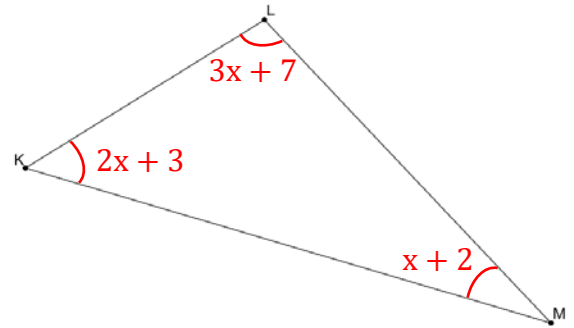
10. Find the perimeter of triangle ABC .

$$\begin{array}{l} P = 2(EF) + 2(DF) + 2(DE) \\ P = 2(9) + 2(8) + 2(6) = 46 \end{array}$$

Use this information to answer questions 1 through 5.

Triangle KLM is shown, where

$m\angle K = (2x + 3)^\circ$, $m\angle L = (3x + 7)^\circ$, and $m\angle M = (x + 2)^\circ$.



1. Find the value of x .

$$(2x + 3) + (3x + 7) + (x + 2) = 180$$

$$6x = 168$$

$$x = 28$$

2. Identify the measure of $m\angle K$.

$$m\angle K = 2(28) + 3 = 59^\circ$$

3. Identify the measure of $m\angle L$.

$$m\angle L = 3(28) + 7 = 91^\circ$$

4. Identify the measure of $m\angle M$.

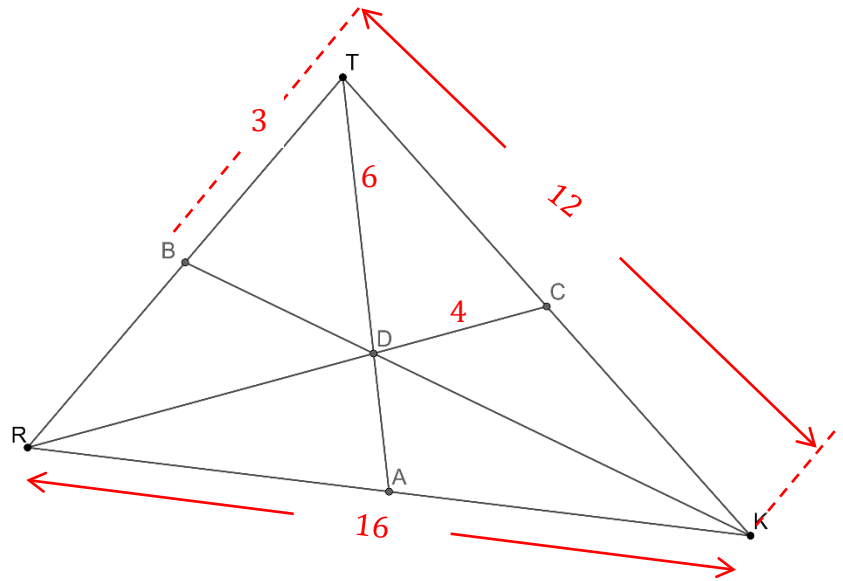
$$m\angle M = 28 + 2 = 30^\circ$$

5. List the sides of the triangle in order of their lengths, smallest to largest.

$\overline{KL}$, $\overline{LM}$, $\overline{KM}$

Use this information to answer questions 6 through 10.

Triangle RTK is shown, where $\overline{RC}$, $\overline{BK}$, and $\overline{AT}$ are median. $BT = 3$, $BK = 12$, $DC = 4$, $TD = 6$, and $RK = 16$.



6. Find the length of $\overline{BD}$.

$$BD = \frac{1}{3}(BK) = \frac{1}{3}(12) = 4$$

7. Find the length of $\overline{RC}$.

$$RC = 3(DC) = 3(4) = 12$$

8. Find the length of $\overline{DA}$.

$$DA = \frac{1}{2}(TD) = \frac{1}{2}(6) = 3$$

9. Find the length of $\overline{RT}$.

$$RT = 2(BT) = 2(3) = 6$$

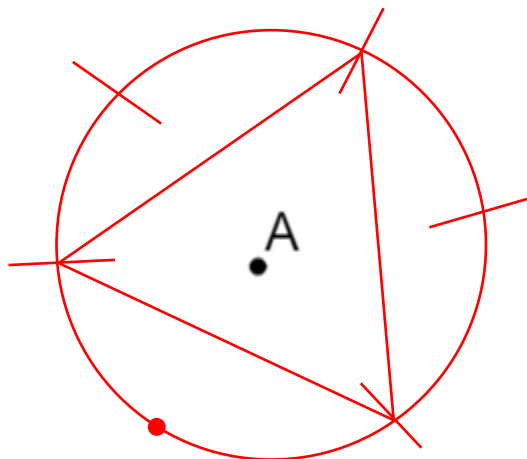
10. Find the perimeter of triangle RBD .

$$\begin{aligned} P &= RB + BD + RD \\ &= 3 + 4 + 8 = 15 \end{aligned}$$

Complete the missing steps showing one way to construct an equilateral triangle inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	Preserve the distance on your compass from the center of the circle to any point on the circle.
3	Mark any point on the circumference of the circle.
4	Place the point of the compass on the point on the circle and make a mark where the compass intersects the circle.
5	Continue around the circle making a total of five small arcs.
6	Mark points where the arcs intersected the circle.
7	Connect every other point to create an equilateral triangle.

Use Point A to construct an equilateral triangle inscribed inside a circle.



Complete the missing steps showing one way to construct a hexagon inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	Preserve the distance on your compass from the center of the circle to any point on the circle.
3	Mark any point on the circumference of the circle.
4	Place the point of the compass on the point on the circle and make an arc where the compass intersects the circle.
5	Continue around the circle making a total of five small arcs.
6	Mark points where the arcs intersect the circle.
7	Connect adjacent points to create a hexagon.

Use Point *P* to construct a hexagon inscribed inside a circle.

